

Free Energy Is a Signed, Zero-Training Detector of Acquisition Shift in Frozen Vision–Language Encoders: A Pre-Registered Falsification Study on Medical Image Corruptions

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Abstract

Frozen vision–language encoders such as CLIP enable zero-shot medical image classification in which class anchors are text embeddings and the softmax posterior over image–anchor distances is exactly a Gibbs distribution at temperature $T = 2\tau$, where τ is the temperature learned during contrastive pre-training. This identification makes two label-free observables available at no training cost: the Shannon entropy $\mathcal{H}$ of the posterior and the Helmholtz free energy F of the underlying energy model. Because a uniform translation of the anchor distances leaves $\mathcal{H}$ invariant but shifts F , we pre-registered a falsifiable prediction: under increasing acquisition shift, F separates source from corrupted images (AUROC ≥ 0.70) while $\mathcal{H}$ stays near chance (AUROC ≤ 0.60). We tested this prediction with a strictly zero-training protocol—prompts frozen before any target evaluation, temperature fixed at 2τ , every reported number traceable to a versioned run—on the official MedMNIST-C corruptions of BreastMNIST (ultrasound, seven corruptions) and PneumoniaMNIST (chest X-ray, four corruptions), with paired DeLong tests and stratified bootstrap confidence intervals. The strong form of the prediction is *refuted*: entropy is never near chance. What the data reveal instead is sharper. On structural corruptions (pixelation, JPEG, motion blur, speckle) F dominates $\mathcal{H}$ significantly (severity-5 AUROC up to 1.000 vs. 0.888; paired Δ AUROC $+0.080$ to $+0.208$, all $p < 0.05$), whereas photometric corruptions (brightness, contrast, X-ray Gaussian noise) *decrease* F and drive the one-sided detector below or near chance (AUROC 0.127–0.594): free energy is a *signed* shift score. A two-sided variant $|F - \text{med}_{\text{source}}(F)|$, calibrated on source images only, restores above-chance detection on five of six sign-inverted cases (e.g., $0.127 \rightarrow 0.803$) while preserving the structural ones. Across prompt reformulations F is markedly more stable than $\mathcal{H}$ (severity-5 AUROC dispersion 0.03 vs. 0.08). All code, per-run metrics and raw scores are released; refuted criteria are reported as pre-registered.

1 Introduction

Zero-shot classification with frozen vision–language encoders (Radford et al., 2021) is an attractive substrate for medical imaging: class definitions are text prompts, no gradient touches the backbone, and the deployment surface is a single forward pass. Yet clinical images arrive from scanners, probes and acquisition protocols that drift over time and across sites, and a deployed zero-shot pipeline needs a label-free signal that the input distribution has moved (Rabanser et al., 2019). Such a signal is the gating quantity for every downstream mechanism studied in test-time adaptation (TTA) (Wang et al., 2021; Shu et al., 2022; Yoon et al., 2024; Niu et al., 2022, 2023): whether to adapt, when to stop, and which samples to refer to a human reader.

This paper studies the cheapest conceivable such signal. When image embeddings $\mathbf{z}$ are scored against K fixed text anchors $\boldsymbol{\mu}_k$, the CLIP posterior is a Gibbs (Boltzmann) distribution over squared distances at temperature $T = 2\tau$ (Section 2). Two observables then come for free: the prediction entropy $\mathcal{H}$ —the workhorse of TTA objectives and confidence baselines (Hendrycks and Gimpel, 2017; Wang et al., 2021)—and the Helmholtz free energy F , the log-partition function popularized for out-of-distribution detection on classifier logits (Liu et al., 2020; Grathwohl et al., 2020; LeCun et al., 2006) and later transposed to vision–language models (Ming et al., 2022).

Our starting point is a structural asymmetry between the two. If an acquisition shift moves images away from *all* anchors roughly uniformly — adding a constant c to every energy E_k — the posterior and hence $\mathcal{H}$ are unchanged, but F shifts by exactly c . This motivated a *pre-registered, falsifiable* prediction (criteria fixed before any target data were scored, Section 4): under increasing corruption severity, F detects the shift (AUROC ≥ 0.70) while $\mathcal{H}$ remains near chance (AUROC ≤ 0.60), with a margin of at least 0.05.

We tested this prediction under a deliberately austere protocol: no training, no adaptation, prompts frozen *a priori* and published verbatim, temperature fixed at the theoretical value 2τ , and every number in this paper generated by a versioned script and stored in a per-run metrics file. The testbed is MedMNIST-C (Di Salvo et al., 2024), the corruption benchmark built on MedMNIST v2 (Yang et al., 2023), on two modalities: breast ultrasound (BreastMNIST, derived from BUSI (Al-Dhabyani et al., 2020)) and chest X-ray (PneumoniaMNIST, derived from Kermany et al., 2018).

Findings. The strong form of the prediction is *refuted*: $\mathcal{H}$ is never near chance (severity-5 AUROC 0.462–0.888), because realistic corruptions do not translate energies uniformly. The refutation, however, uncovers a structure that we believe is more useful than the original conjecture:

1. **A structural/photometric dichotomy.** On corruptions that destroy spatial structure (pixelation, JPEG compression, motion blur, speckle noise), F dominates $\mathcal{H}$ significantly as a shift detector. On photometric corruptions (brightness, contrast, X-ray Gaussian noise), corrupted embeddings move *closer* to the text anchors, F decreases, and the one-sided detector drops below chance — to 0.127 in the worst case. Free energy is a *signed* score, not a magnitude.
2. **A source-calibrated two-sided rescue.** The statistic $|F - \text{med}_{\text{src}}(F)|$, using only the source median, restores above-chance detection on five of the six sign-inverted cases while leaving structural detection essentially intact.
3. **Prompt stability.** Across three prompt reformulations, the severity-5 detection AUROC of F varies by ~ 0.03 while that of $\mathcal{H}$ varies by ~ 0.08 : the partition function is more robust to the wording that defines the anchors than the posterior shape is.

We report the pre-registered verdict in full, including the criteria that failed, and release code, metrics and raw scores for every run.

2 Theoretical framework

2.1 Gibbs posterior over textual anchors

Let $\mathbf{z} \in \mathbb{R}^d$ be an ℓ_2 -normalized image embedding and $\boldsymbol{\mu}_1, \dots, \boldsymbol{\mu}_K$ the ℓ_2 -normalized text embeddings of the class prompts. Define the configuration energy

$$E_k(\mathbf{z}) = \|\mathbf{z} - \boldsymbol{\mu}_k\|_2^2 = 2 - 2\langle \mathbf{z}, \boldsymbol{\mu}_k \rangle. \quad (1)$$

The Gibbs distribution at temperature $T > 0$ is $p_k(\mathbf{z}) = \exp(-E_k/T) / \sum_j \exp(-E_j/T)$. CLIP computes logits $\langle \mathbf{z}, \boldsymbol{\mu}_k \rangle / \tau$ with a learned temperature τ ; substituting (1) shows the two softmaxes coincide exactly when

$$T = 2\tau. \quad (2)$$

No temperature is tuned in this work: T is read off the frozen model (here $T \approx 0.0200$).

2.2 Observables and the thermodynamic identity

We track the entropy, free energy, mean energy and a normalized order parameter (“crystallinity”):

$$\mathcal{H}(\mathbf{z}) = -\sum_k p_k \log p_k, \quad F(\mathbf{z}) = -T \log \sum_k e^{-E_k/T}, \quad \langle E \rangle(\mathbf{z}) = \sum_k p_k E_k, \quad (3)$$

$$\chi(\mathbf{z}) = 1 - \mathcal{H}(\mathbf{z}) / \log K, \quad \text{with the identity} \quad F = \langle E \rangle - T \mathcal{H} \quad (4)$$

used as a software invariant (unit-tested to numerical precision). F is, up to the distance parameterization, the negative energy score of Liu et al. (2020) evaluated on distance logits.

2.3 The pre-registered prediction

Proposition 1 (Translation asymmetry). *For any constant c , the map $E_k \mapsto E_k + c$ leaves p and therefore $\mathcal{H}$ and χ unchanged, while $F \mapsto F + c$.*

If acquisition shift pushed embeddings away from all anchors while roughly preserving relative class separations, F would detect it and $\mathcal{H}$ would not. Before scoring any corrupted image we therefore froze the following decision criteria, evaluated at maximum severity with the source cohort as negatives:

- C1. $\text{AUROC}(F) \geq 0.70$;
- C2. $\text{AUROC}(\mathcal{H}) \leq 0.60$;
- C3. $\text{AUROC}(F) - \text{AUROC}(\mathcal{H}) \geq 0.05$.

The prediction is *confirmed* only if all three hold. These thresholds are decision rules, not results; a cleanly executed refutation is reported as such.

3 Related work

Energy-based OOD detection. Liu et al. (2020) showed that the logit log-partition function outperforms maximum softmax probability (Hendrycks and Gimpel, 2017) for OOD detection; Grathwohl et al. (2020) read classifiers as energy-based models. Ming et al. (2022) transposed confidence-based OOD scores to CLIP. Our contribution is orthogonal to score engineering: a pre-registered, severity-resolved comparison of F against $\mathcal{H}$ on *medical acquisition-style* corruptions, exposing the signed behavior of F that global OOD benchmarks average away.

Dataset-shift detection. Rabanser et al. (2019) compared two-sample tests on learned representations. We restrict attention to the two statistics that are free in a deployed zero-shot pipeline, and quantify their per-sample discriminative power (AUROC) rather than cohort-level test power.

Test-time adaptation. Entropy minimization is the dominant TTA objective (Wang et al., 2021; Niu et al., 2022, 2023), extended to prompt tuning in vision-language models (Shu et al., 2022) with calibration corrections (Yoon et al., 2024). Our results bear on the *gating* of such methods: a signal that moves under shift while entropy barely moves (or moves for the wrong reason) is the natural trigger and stopping observable. This paper establishes the zero-training baseline for that program; the adaptation stage itself is out of scope here.

Benchmarks. MedMNIST-C (Di Salvo et al., 2024) provides modality-specific corruptions at five severities for MedMNIST v2 (Yang et al., 2023); we use its official corruption operators verbatim.

4 Experimental protocol

4.1 Substrate

Encoder: CLIP ViT-B/16 (OpenAI weights, QuickGELU variant) via `open_clip` (Ilharco et al., 2021), entirely frozen, run in float32 (PyTorch 2.4.1, Python 3.11, Windows). Most runs were executed on CPU; the three PneumoniaMNIST runs were re-executed on an RTX 4060 (CUDA 12.4 build of the same PyTorch version) after a repository-hygiene fix. Embeddings agree across the two devices to a cosine similarity above 0.999998 and the reported AUROCs to 3×10^{-4} ; each run records its own device. Temperature fixed at $T = 2\tau \approx 0.0200$ per (2). Anchors are the ℓ_2 -normalized means of the per-class prompt-template embeddings (prompt ensembling), renormalized after averaging.

4.2 Cohorts and corruptions

BreastMNIST (Yang et al., 2023; Al-Dhabyani et al., 2020): breast ultrasound, binary malignant vs. normal/benign, test split $n = 156$ at 224×224 (MedMNIST+). All seven official MedMNIST-C corruptions for this dataset: `speckle_noise`, `motion_blur`, `brightness_up`, `brightness_down`, `contrast_down`, `jpeg_compression`, `pixelate`, at severities 0 (clean source) through 5. **PneumoniaMNIST** (Kermany et al., 2018): chest X-ray, binary normal vs. pneumonia, test split $n = 624$; corruptions `gaussian_noise`, `gaussian_blur`, `brightness_down`, `contrast_down`.

Zero-shot diagnostic performance on clean data is deliberately reported and modest — BreastMNIST balanced accuracy 0.517 (near chance; accuracy 0.712, AUROC 0.636), PneumoniaMNIST accuracy 0.620 — the shift detection question is independent of diagnostic quality, and no claim of clinical utility of the zero-shot classifier is made.

4.3 Prompts, frozen a priori

The prompt sets were fixed before any corrupted image was scored and are reproduced verbatim. BreastMNIST classes {“a malignant breast tumor”, “a normal or benign breast tissue”} with templates {“a breast ultrasound image of {}.”, “an ultrasound scan showing {}.”, “grayscale sonography of {}.”}; PneumoniaMNIST classes {“a normal chest”, “a chest with pneumonia”} with templates {“a chest x-ray image of {}.”, “a radiograph showing {}.”}. Prompt sensitivity is measured by re-running the full protocol with each single template (Section 5.4).

4.4 Metrics and statistics

For each severity $s \geq 1$ we compute the AUROC of F (and of $\mathcal{H}$) as a per-sample discriminator between the clean cohort ($s = 0$, negatives) and the corrupted cohort (positives); higher score should indicate the corrupted domain. Because both scores are computed on the same samples,

Table 1: Severity-5 shift detection (source vs. corrupted), canonical runs (prompt ensemble, seed 0, severities 0–5). $\Delta = \text{AUROC}(F) - \text{AUROC}(\mathcal{H})$ with stratified paired bootstrap 95% CI and paired DeLong p -value. $|F|_2 = \text{two-sided variant } |F - \text{med}_{\text{src}}(F)|$. Structural corruptions above the rule, photometric below. Run identifiers, in table order: 3e02269f, 98d42003, 2308a744, de6dba0c, 2bc1d5f9, 41fa7739, c1697982, 4ae2e534, a0dfe1a7, 2de86cbd, 2764835a; each is the short hash of a `metrics.json` released with the code.

Dataset	Corruption	AUROC(F)	AUROC($\mathcal{H}$)	Δ [95% CI]	p	$ F _2$
Breast	pixelate	1.000	0.888	+0.112 [+0.078, +0.150]	1.9×10^{-9}	1.000
Breast	jpeg_compression	0.920	0.711	+0.208 [+0.143, +0.274]	7.7×10^{-10}	0.899
Breast	motion_blur	0.885	0.760	+0.125 [+0.060, +0.190]	1.7×10^{-4}	0.845
Breast	speckle_noise	0.814	0.734	+0.080 [+0.006, +0.154]	0.032	0.742
Pneumonia	gaussian_blur	0.970	0.798	+0.172 [+0.148, +0.198]	4.1×10^{-41}	0.962
Breast	brightness_up	0.594	0.677	−0.083 [−0.176, +0.012]	0.083	0.447
Breast	brightness_down	0.359	0.563	−0.204 [−0.297, −0.112]	1.3×10^{-5}	0.606
Breast	contrast_down	0.299	0.498	−0.199 [−0.287, −0.112]	7.4×10^{-6}	0.630
Pneumonia	gaussian_noise	0.127	0.462	−0.335 [−0.368, −0.300]	1.7×10^{-86}	0.803
Pneumonia	brightness_down	0.200	0.644	−0.445 [−0.482, −0.406]	8.0×10^{-118}	0.598
Pneumonia	contrast_down	0.209	0.698	−0.490 [−0.526, −0.452]	1.7×10^{-148}	0.589

we compare them with the paired DeLong test (DeLong et al., 1988) and a stratified paired bootstrap (Efron and Tibshirani, 1994) with 10^4 resamples for 95% confidence intervals on $\Delta \text{AUROC} = \text{AUROC}(F) - \text{AUROC}(\mathcal{H})$. The two-sided variant replaces a sample’s score x by $|x - \text{med}_{\text{src}}|$ where med_{src} is the median of the *source* scores; no target information enters the calibration.

4.5 Leakage controls and traceability

(i) Prompts frozen before target evaluation; (ii) $T = 2\tau$, never tuned; (iii) no threshold is fit on target data (the two-sided center uses source only); (iv) every number below is traceable to a run identifier listed in the caption of Table 1; each run stores its configuration, git commit, code hash, environment, per-severity metrics and raw per-sample scores. A caveat we cannot close: CLIP’s web-scale pre-training corpus is not auditable, so memorization of the public source images cannot be excluded; corruption operators, however, are applied locally and cannot have been seen.

5 Results

5.1 Pre-registered verdict: the strong prediction is refuted

Table 1 reports severity-5 detection for all canonical runs. Criterion C2 — entropy near chance — fails on *every* corruption (minimum severity-5 AUROC($\mathcal{H}$) = 0.462, maximum 0.888). The conjunction $C1 \wedge C2 \wedge C3$ is therefore refuted in all configurations, as recorded in each run’s verdict block (`prediction_confirmee: false`). Realistic corruptions do not translate the energy landscape uniformly: they also reshape relative distances, which entropy sees.

5.2 Free energy is a signed detector

The refutation has a clean geometry. On structural corruptions the corrupted embeddings move *away* from all text anchors: mean F rises monotonically with severity (BreastMNIST speckle: $1.288 \rightarrow 1.351$; pixelate: $1.288 \rightarrow 1.479$) and F dominates $\mathcal{H}$ significantly (Table 1, upper block; all DeLong $p < 0.05$, CIs excluding zero). On photometric corruptions the embeddings move

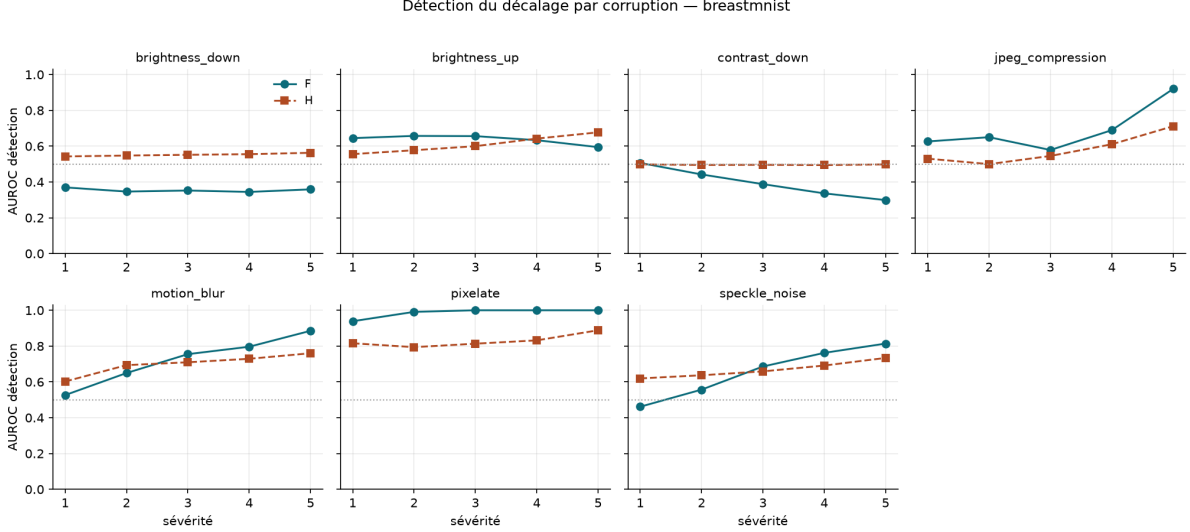


Figure 1: Severity-resolved detection AUROC of F (solid) and $\mathcal{H}$ (dashed) for the seven official BreastMNIST-C corruptions. Dotted line: chance. Note the below-chance trajectories of F on `brightness_down` and `contrast_down`.

closer to the anchors — mean F decreases (`contrast_down`: $1.288 \rightarrow 1.266$; PneumoniaMNIST `gaussian_noise`: $1.264 \rightarrow 1.234$) — and the one-sided AUROC falls below 0.5, to 0.127 in the extreme case. A plausible reading is that brightness/contrast compression and moderate X-ray noise push images toward the generic, high-density region of CLIP’s image manifold, reducing the image–text modality gap; we do not claim a mechanism beyond the observed sign.

Two practical consequences follow. First, using raw F (or its negation, as in standard energy-score OOD pipelines) as a shift alarm silently fails on exactly the photometric drifts most typical of scanner and probe settings. Second, the failure is directional, hence recoverable.

5.3 A source-calibrated two-sided score rescues the inversions

The statistic $|F - \text{med}_{\text{src}}(F)|$ uses only source images. At severity 5 it restores above-chance detection on five of the six sign-inverted cases — PneumoniaMNIST `gaussian_noise` $0.127 \rightarrow 0.803$, BreastMNIST `contrast_down` $0.299 \rightarrow 0.630$, `brightness_down` $0.359 \rightarrow 0.606$, PneumoniaMNIST `brightness_down` $0.200 \rightarrow 0.598$ and `contrast_down` $0.209 \rightarrow 0.589$ — while leaving structural cases essentially unchanged (`pixelate` 1.000, `jpeg` 0.899, `motion_blur` 0.845, `gaussian_blur` 0.962). The recovery is strong where the downward drift is large (X-ray Gaussian noise) and modest where the drift is shallow relative to source dispersion (X-ray photometric shifts, ≈ 0.59 , where $\mathcal{H}$ itself reaches 0.64–0.70). The residual failure, `brightness_up` (0.447), is a genuinely hard case: its F trajectory is non-monotonic (up at low severity, down at high), so no function of F alone can order it; its entropy signal (0.677) and the X-ray photometric cases together suggest a bivariate $(F, \mathcal{H})$ score as the natural next step.

5.4 Prompt sensitivity: F is more stable than $\mathcal{H}$

The anchors are literally defined by prompt wording, so prompt sensitivity is the substrate-specific confound. Re-running the full protocol with each single template (BreastMNIST, severity 5):

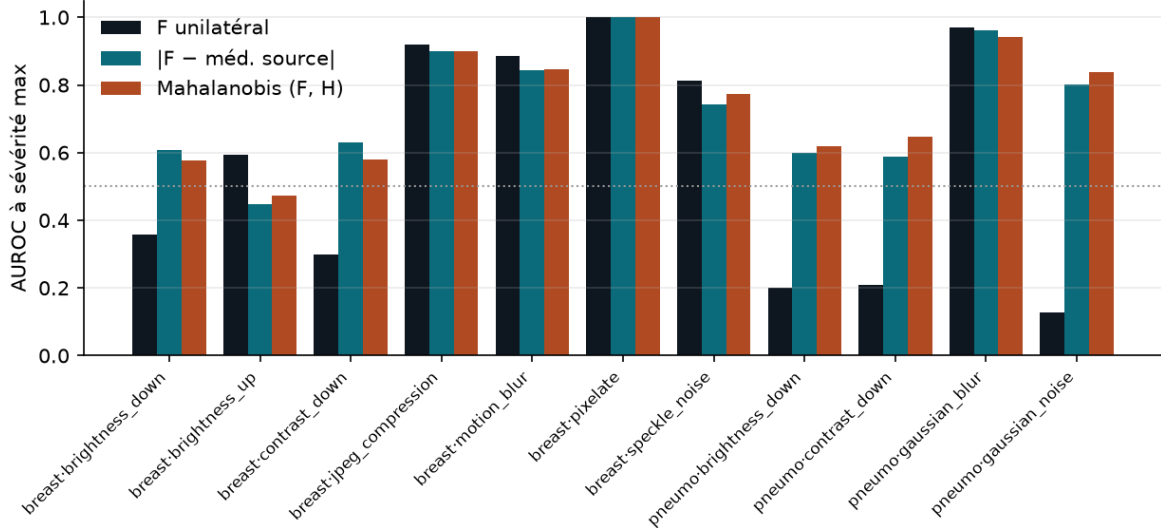


Figure 2: One-sided F vs. two-sided $|F - \text{med}_{\text{src}}(F)|$ detection AUROC at maximum severity, all canonical runs.

Corruption	AUROC(F)				AUROC($\mathcal{H}$)			
	t0	t1	t2	ens.	t0	t1	t2	ens.
speckle_noise	0.778	0.801	0.854	0.814	0.777	0.641	0.597	0.734
brightness_down	0.351	0.419	0.307	0.359	0.580	0.464	0.609	0.563

The dispersion of F across templates (population s.d. over the three isolated templates: 0.032 on speckle, 0.046 on brightness_down) is roughly half that of $\mathcal{H}$ (0.076 and 0.062): the partition function aggregates the anchor geometry and is less exposed to a particular wording than the posterior shape. The sign inversion on brightness_down persists under *every* template — it is not a prompt artifact.

5.5 Robustness to corruption sampling

Corruption operators are stochastic. Across three seeds (BreastMNIST, speckle_noise, severity 5): $\text{AUROC}(F) = 0.814/0.819/0.827$, $\text{AUROC}(\mathcal{H}) = 0.734/0.749/0.738$; the paired severity-5 ΔAUROC ranges +0.069 to +0.088 with overlapping CIs. Sampling noise is an order of magnitude smaller than the effects reported above.

5.6 Second modality

The chest X-ray cohort replicates the dichotomy on an independent modality, anatomy and corruption implementation, with $4\times$ larger test sets ($n = 624$). Gaussian blur (structural) behaves like the ultrasound structural block: F rises monotonically ($1.264 \rightarrow 1.320$) and dominates $\mathcal{H}$ (0.970 vs. 0.798, $\Delta = +0.172$, $p = 4.1 \times 10^{-41}$). All three photometric corruptions invert the sign of F — Gaussian noise, brightness and contrast compression each move X-ray embeddings *closer* to the text anchors (F AUROC 0.127–0.209), with effect sizes far beyond sampling noise ($|\Delta| \geq 0.335$, $p \leq 10^{-86}$). On this modality the inversions are deeper than in ultrasound, and $\mathcal{H}$ is the better detector for the two photometric-compression cases (0.644 and 0.698): neither observable subsumes the other, which is precisely the argument for reporting both and for the bivariate score sketched above.

6 Discussion and limitations

What the refutation teaches. The pre-registered conjecture assumed acquisition shift acts as a quasi-uniform translation of the energy landscape. The data say otherwise: realistic corruptions move embeddings along the modality-gap axis in *either* direction and simultaneously reshape relative distances. The corrected picture — F as a signed coordinate of drift along the anchor-distance axis, $\mathcal{H}$ as a shape statistic — is more useful than the original conjecture, and directly actionable via the two-sided score.

Implications for test-time adaptation. Entropy-minimizing TTA (Wang et al., 2021; Shu et al., 2022) adapts wherever entropy is high. Our results exhibit shifts where entropy barely moves while F moves strongly (JPEG: $\mathcal{H}$ AUROC 0.711 vs. F 0.920) and shifts where entropy rises while F falls. A gating rule based on $|F - \text{med}_{\text{src}}|$ is the zero-cost candidate this study puts forward for the adaptation-triggering and early-stopping stages of our broader research program on frozen vision-language encoders in ultrasound.

Limitations. (i) Test cohorts are small ($n = 156$ ultrasound images); CIs are reported accordingly. (ii) Single backbone (CLIP ViT-B/16, OpenAI weights); a biomedical encoder (Zhang et al., 2023) may shift the geometry and is the designated robustness check. (iii) MedMNIST-C corruptions are synthetic surrogates for acquisition shift; the multi-center ultrasound transfer (BUSI→UDIAT→BUS-UCLM) is the pre-registered next cohort. (iv) The two-sided median is computed on the full source cohort (in-sample); with $n = 156$ and one degree of freedom the optimism is negligible, but a held-out calibration split is the clean design. (v) Images are 224×224 upsamplings of MedMNIST sources. (vi) CLIP pre-training contamination cannot be audited (Section 4).

7 Conclusion

Under a zero-training, leakage-controlled, pre-registered protocol on official medical corruption benchmarks, the Helmholtz free energy of a frozen CLIP posterior is a strong but *signed* detector of acquisition shift: significantly better than entropy on structural corruptions, below chance on photometric ones, and repaired in five of six failure cases by a two-sided source-calibrated variant that requires nothing but source images. The strong form of our pre-registered prediction — entropy at chance — is refuted and reported as such. Every number in this paper is traceable to a versioned run; code, metrics and raw scores are released with the paper.

Reproducibility. No training anywhere; ~ 3 minutes per corruption sweep on CPU (156 images $\times$ 6 severities), ~ 5 seconds on the GPU. Each `metrics.json` records the device it ran on. Code, per-run `metrics.json`, raw scores and analysis scripts are available in the project repository.

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